

Computational Screening of Redox-Active Monomers

A physics-based, multi-objective candidate-selection pipeline

RTFB Materials Research

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Redox potential · Stability · Kinetics · Capacity

System. A **Merrifield-resin polymer** carrying pendant redox groups. The redox chemistry is *local*, so we model the **monomer as a proxy**: establish we can capture the local chemistry, then add backbone effects.

Selection is multi-objective. Voltage and stability are necessary but not sufficient:

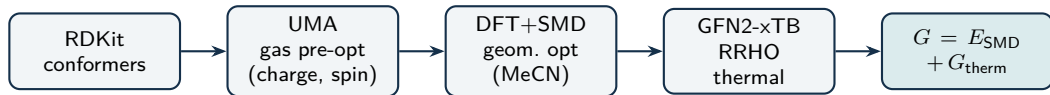
$$\text{Energy density} \propto n \cdot V_{\text{cell}} \cdot \text{solubility}$$

- **redox potential** $\rightarrow$ cell voltage / window
- **stability** $\rightarrow$ cycle life
- **kinetics** (λ), **capacity** (n), cost, crossover

Guiding principle

Energy *differences* with error cancellation compute **reliably**; *absolute* charged-species solvation does **not**.

Build objectives on the cancellation-friendly quantities; treat the rest as labelled proxies or filters.



Level of theory

- opt: r²SCAN/def2-SVP(D) + SMD
- energy: ω B97M-V/def2-TZVP
- referenced to level-matched Fc/Fc⁺

Two hard-won fixes

- gpu4pyscf open-shell Hessian is **broken** → thermal from **GFN2-xTB** RRHO
- cross-points gas-only (do_smd=False), ~2× cheaper

Blind test on OROP MeCN potentials ($n = 36$):

	ours	OROP impl.-DFT
MAE (V)	0.58	0.43
signed bias	+0.46	—
Spearman (all)	0.95	—

Absolute accuracy sits at the **$\sim 0.5\text{--}0.9$ V physics floor** of implicit-solvent DFT — the same for everyone. **Ranking**, not absolute potential, is the usable output.

Ranking must be judged *within* charge class

class	n	MAE	Spearman
cations (+1)	21	0.44	0.885
anions (0)	13	0.77	0.489
anions (filtered)	8	0.53	0.952

Global 0.95 is inflated (charge classes separate).
Cations rank well; anions needed a fix → next slide.

The anion ranking collapse (Spearman 0.49) traced to a few molecules with **no reversible reduction**:

molecule	exp	RMSD _{neu→an}
CCl ₄	-2.57	3.4 Å
CH ₂ Br ₂	-3.12	1.3 Å
(unbound anions)	—	EA _{gas} < 0

Dissociative electron attachment (CCl₄+e⁻→CCl₃[•]+Cl⁻) — scored against irreversible *peak* potentials.

Reversibility screen

Bound anion (solvated EA > 0) *and* intact (**bond-graph unchanged**, not raw RMSD — which is confounded by tether flexibility).

- filtering → anion Spearman **0.49**→**0.95**
- a *required* screening criterion + a *stability* signal
- all 20 candidate couples: **reversible**

Disproportionation $2R \rightarrow R^+ + R^-$
 (atom- & electron-conserving $\rightarrow$ **cancellation**
 $\rightarrow$ **reliable**):

intermediate	ΔG_{disp} (kJ/mol)
TEMPO $\bullet$	+221
AQ semiquinone	+86
viologen MV $^{+\bullet}$	+52

TEMPO — the textbook persistent radical (our validation anchor) — ranks most stable, as it must. Exact from existing energies; no new calculation.

Dimerization $2 \text{MV}^{+\bullet} \rightarrow [\text{MV}_2]^{2+}$ (the real viologen failure mode):

$$\Delta G_{\text{dim}} = +0.49 \text{ eV (flagged)}$$

Config sampling: minimum well-defined ($\sigma_{\text{config}} \approx 2 \text{ meV}$).

The uncertainty is the **+2 pimer desolvation** — the continuum's weak spot.

$\Rightarrow$ use dimerization as a **relative** ranking; lean on disproportionation as the trustworthy axis.

Inner-sphere λ_i via the **4-point (Nelsen)** scheme — *identical* to D3TaLES's definition (verified in their source). Reliable via cancellation; low λ = fast, reversible ET.

couple	λ_i (eV)
ferrocene ox $\rightarrow$ neu	0.085
viologen ox1 $\rightarrow$ neu	0.39
phenothiazine ox $\rightarrow$ neu	0.43
TEMPO ox $\rightarrow$ rad	0.96

Cross-check vs. D3TaLES computed λ

MAD = **0.094 eV** ($n = 3$ overlapping molecules)

e.g. TEMPO hole reorg: ours 0.97 vs 0.94 eV

Our λ implementation is sound. Cross-points computed gas-only for efficiency; geometry-source difference (SMD-opt vs D3TaLES gas-opt) explains the small residual.

What works

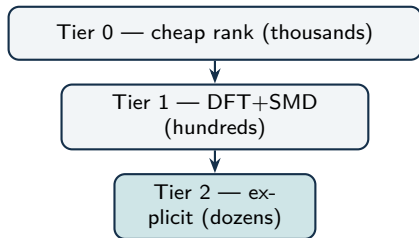
- ranking (within charge class)
- xtb RRHO thermal (open-shell safe)
- disproportionation stability
- reorganization energy λ
- reversibility as filter + stability signal
- capacity (n , MW): exact

Negative results (saved effort)

- continuum ion-pairing: **refuted** (spacing off by ~ 11 V)
- explicit microsolvation (4 MeCN): **no gain** for viologen
- absolute redox potential: capped ~ 0.5 V
- dimerization absolute: **unreliable** (+2 desolvation)
- solubility / crossover: proxies only

Corroborated by the metabolic- ΔG pipeline: 0.14 eV via isodesmic cancellation $\Rightarrow$
cancellation, not more physics, is the lever.

Tiered funnel (explicit solvation is a finisher, never for thousands):



Pareto scorecard — built correctly

- objectives = **trustworthy** axes only (capacity, λ , disprop. stability, ranked potential)
- proxies (solubility, SA, crossover) = **filters**, not axes
- **reversibility** = hard pre-filter
- split **anolyte** / **catholyte** (potential isn't "more is better")
- σ -**aware** domination (don't rank within error)
- pair front with $n \cdot V$ ·solubility figure of merit

1. **Complete the scorecard** — wire in capacity (n , MW), SA cost, and the ΔG_{solv} solubility proxy (all cheap; no new DFT).
2. **Validate the stability axis** against known stable/unstable molecules (as we did for redox vs OROP and λ vs D3TaLES).
3. **Build the selection engine** (pareto.py) — objectives + filters + anolyte/catholyte split + σ -aware domination.
4. **Harden for scale** — SCF-robustness patch ($\sim 20\%$ of jobs hit gradient non-convergence).
5. **Validation** $\rightarrow$ **discovery**: define the Merrifield-monomer design space, run the funnel, produce the **Pareto shortlist**.
6. **Later**: backbone/tether & inter-monomer effects; same-charge referencing for absolute accuracy.

Summary

We can **rank** redox candidates and screen **stability**, **kinetics**, and **reversibility** on a physics-based footing — with each axis **honestly labelled** by how far we can trust it.

Next: assemble the Pareto scorecard and point it at the real design space.